of the Moon in May 1994, the spacecraft failed before it could reach Geographos.)

In principle, you could use big rocket engines, or projectile impact, or equip the asteroid with giant reflective panels and shove it with sunlight or powerful Earth-based lasers. But with technology that exists right now, there are only two ways. First, one or more high-yield nuclear weapons might blast the asteroid or comet into fragments that would disintegrate and atomize on entering the Earth's atmosphere. If the offending worldlet is only weakly held together, perhaps only hundreds of megatons would suffice. Since there is no theoretical upper limit to the explosive yield of a thermonuclear weapon, there seem to be those in the weapons laboratories who consider making bigger bombs not only as a stirring challenge, but also as a way to mute pesky environmentalists by securing a seat for nuclear weapons on the save-the-Earth bandwagon.

Another approach under more serious discussion is less dramatic but still an effective way of maintaining the weapons establishment —a plan to alter the orbit of any errant worldlet by exploding nuclear weapons nearby. The explosions (generally near the asteroid's closest point to the Sun) are arranged to deflect it away from the Earth.* A flurry of low-yield nuclear weapons, each giving a little push in the desired direction, is enough to deflect a medium-sized asteroid with only a few weeks' warning. The method also offers, it is hoped, a way to deal with a suddenly detected long-period comet on imminent collision trajectory with the Earth: The comet would be intercepted with a small asteroid. (Needless to say, this game of celestial billiards is even more difficult and uncertain—and therefore even less practical in the near future—than the herding of an asteroid on a known, well-behaved orbit with months or years at our disposal.)

We don't know what a standoff nuclear explosion would do to an asteroid. The answer may vary from asteroid to asteroid. Some small worlds might be strongly held together; others might be little more than self-gravitating gravel heaps. If an explosion breaks, let's say, a 10-kilometer asteroid up into hundreds of 1-kilometer fragments,